

Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-5807
Project Name	AI-STUDYBUDDY
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	[e.g., JMeter, Locust, k6, Postman]
Type of Testing	[e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	AI endpoints: POST /api/materials/:id/summarize, /flashcards, /quiz, /study-plan; POST /api/auth/login
Test Environment	[e.g., Local / Cloud / Hosted Server]
Test Date	

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1				
2				
3				
4				

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds			
2	Response Time (Max)	< 5 seconds			
3	Throughput (Req/sec)				
4	Error Rate	< 1%			
5	CPU Utilization	< 80%			
6	Memory Utilization	< 80%			

Step 4: Observations & Analysis

Key Findings:
 [Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
 [List any performance bottlenecks observed during testing]

Optimization Steps Taken:
 [Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]